COMP1800 – Data Visualization

2025-26 - Term 2

Coursework Report

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MSc. Data Science

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1. Introduction to Data Visualization

Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics and even animations to communicate complex data relationships and data-driven insights in a way that is easy to understand (IBM, 2021). It leverages the human visual system's ability to identify shapes and colors, allowing users to process complex datasets more efficiently than traditional spreadsheets or text-based reports. An effective data visualization is not merely about creating aesthetically pleasing graphics; it is about forging a direct path from data to insight. Its primary goal is to provide the viewer with the greatest number of ideas in the shortest time with the least ink in the smallest space (Tufte, 2001).

This report aims to present the findings obtained after exploring ChrisCo company's 2023 venue data. A highly structured, narrative-driven data exploration was carried out where the flow starts off from a high-level overview of visitor volumes to time-series trends where some possible anomalies among the data was identified, seasonality, and finally into multi-dimensional relationships between the venue attributes and also provisioning a deep dive into one of the strong relationships.

2. Visualizations and Insights

2.1. Sorted and Segmented Bar Chart for Visitor Volume Across Different Venues

Figure 1 provides a high-level overview of the data as a whole. It acts as the foundation for all the visuals that follow, allowing the reader to immediately understand which venues dominate and which are underperforming. Most importantly, through the sorted bars, the natural gaps among the venues are identified justifying the segmentation thresholds, making it feel data-driven rather than random assignment. This segmentation is then further used for the rest of the analysis as well. This visualization uses the code identity from “L02 Proportion / 07BarChart all coloured.py and 08BarChart all sorted.py”. To adapt to this dataset, aspects of both examples were combined so they were sorted and then assigned colors for different segments and limit lines drawn as well to enhance readability.

Figure 1 shows clear skewed distribution among the 35 venues, four high-volume venues UOR, VUJ, VAU and PFN clearly stand out with total visitors more than 150,000, 8 medium with visitors in-between 50,000 and 100,000, and remaining as low-volume venues. To visually convey the hierarchical information, green color represents top performers, yellow shows mid-tier and finally red invokes a visual message that these are low performers. Also, the visual shows that there might be some anomalies which are underperforming a lot and might need further attention.

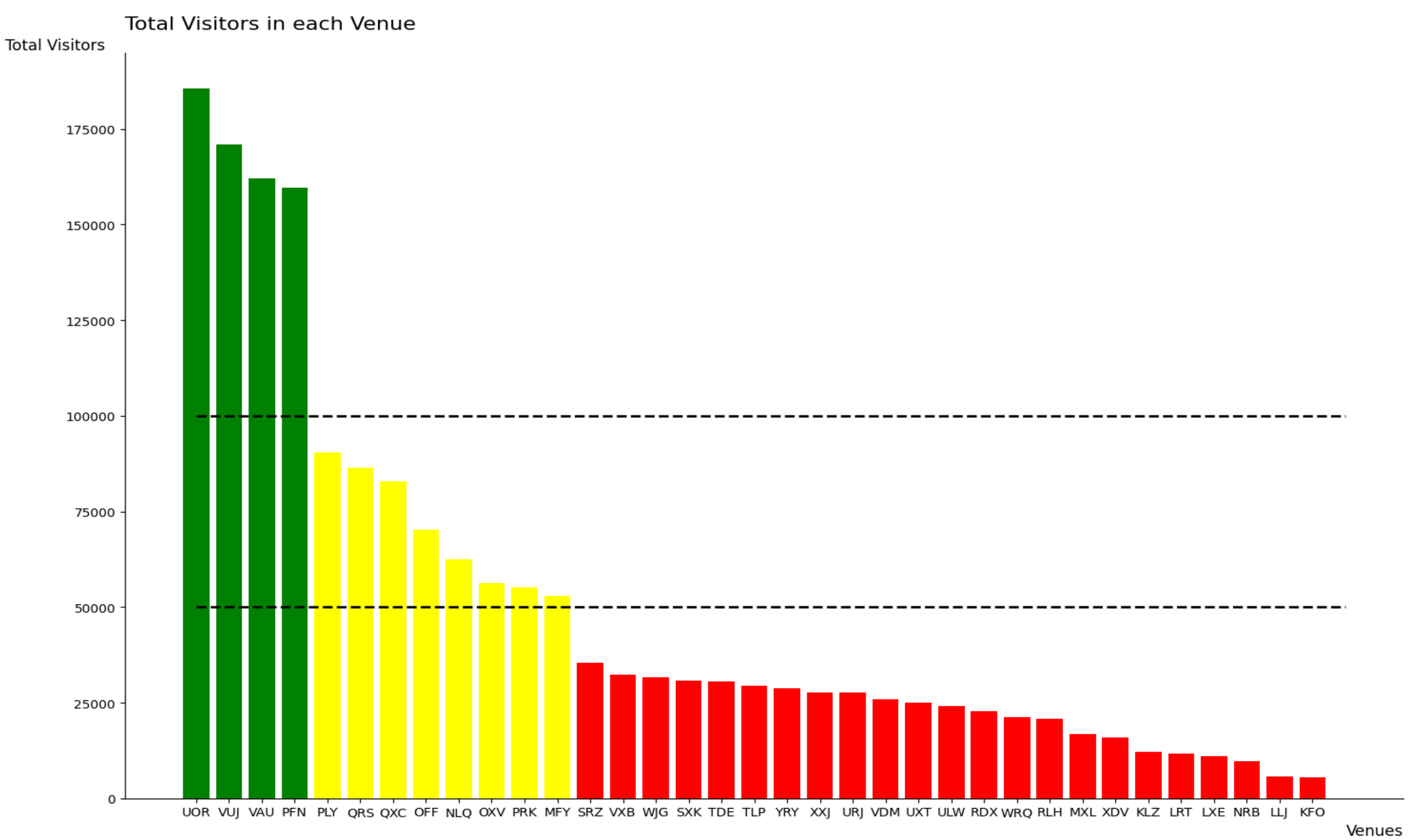

Figure 1: Total visitors across different venues

2.2. Distribution Plot for All Venues

If Figure 1, gave high-level overview of the whole data, Figure 2 narrows it down to specific venues showcasing visitor distribution on a daily basis. It uses multiple histogram subplots to describe how the daily visitors are distributed among different venues independently. It shows where the data normally groups up and where the outliers are. This makes it easier to identify patterns like variability, skewness, and demand consistency. It is important because the venues showing concerning fluctuations and low performance can be isolated for deeper exploration. This visualization's code identity is from "L05 Distribution / 02Histogram all subplot.py".

Figure 2 reveals that even though most of the distribution are normal, there are several right-skewed distributions at venues LLJ, KLZ, MXL, XDV, KFO, LXE, NRB and LRT which means that most of the days they had low to moderate visitors, and few days they experienced high spikes as well. This can be interpreted as possible events like maybe weekends or festive events which might have occasionally surged the visitors so the volume is always unpredictable. They can be confirmed to be anomalies. For the venues that showcase symmetric distribution, there performance is consistent and predictable allowing good basis for forecasting surges or sparsity in visitors.

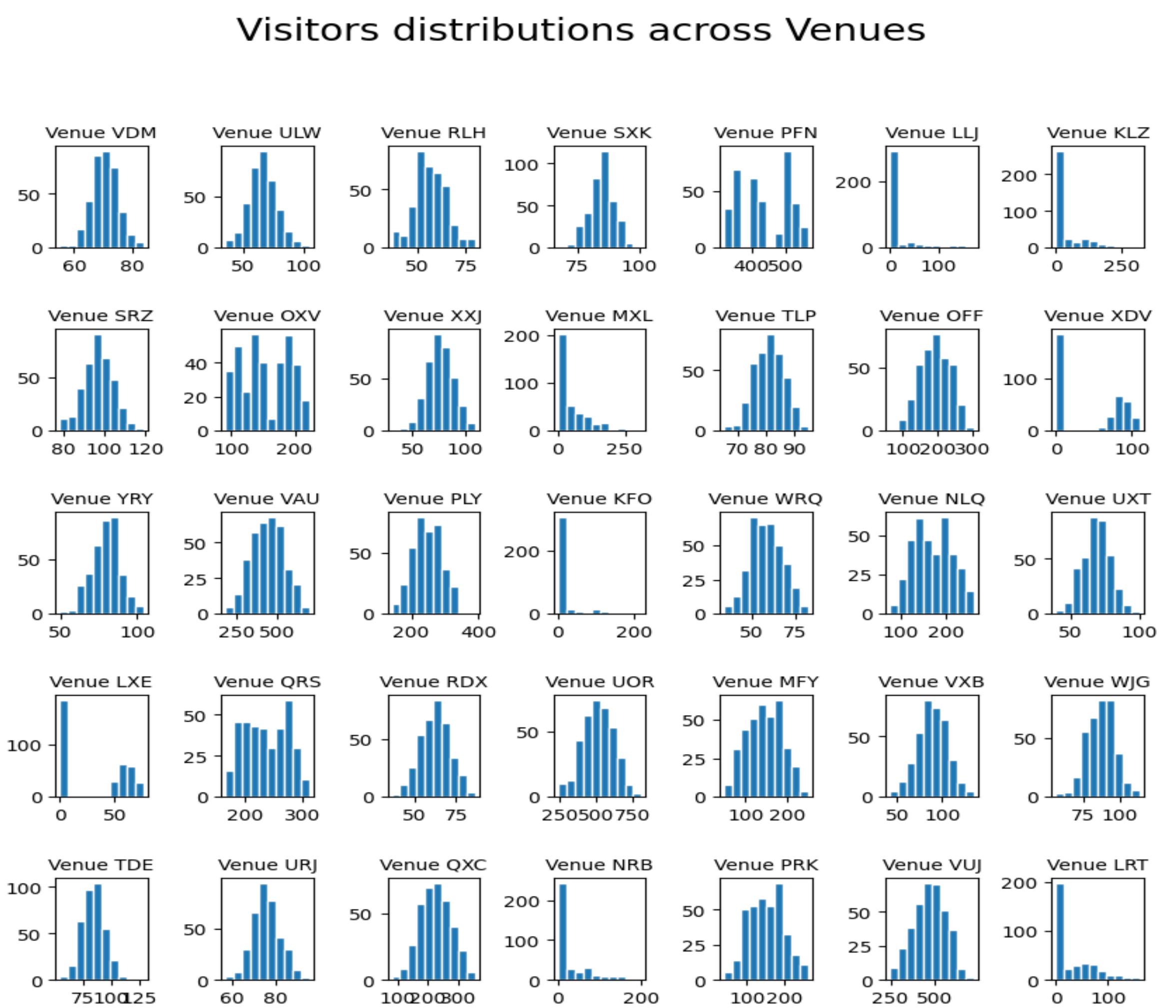

Figure 2: Visitor distribution across the venues

2.3. Interactive Line Plot to Investigate Anomalies

Following the confirmation of anomalies among the venues, the next step is to investigate those anomalies. Although, the company prioritizes high and medium volume venues, it still needs a clear idea of the low-volume anomalous venues and the reason behind their occurrence. This helps the company to plan and take necessary steps to mitigate such issues in the future and work on their growth. That's why, an interactive line plot makes sense as it provides the reader with perks of exploring and dissecting the data through the interactive tools provided, so they can look into the ones they are concerned about. The line plot is rolling averaged to 28 days to smooth out noises to allow the reader clearly observe upward or downward trends across the year. Code identity is from "L07Interaction / 02LinePlot high volume.py and L03Time / 05LinePlot high volume rolling averaged.py". Since, it was interactive line plot but rolling averaged, the code had to be adapted to combine both by using the '*' operator in hvPlot which allows multiple plots to overlay on top of each other. Furthermore, if the 8 overlapping anomalous venues look too chaotic, the reader can just click on the interactive legend to mute or un-mute specific lines dynamically, making visual exploration far more effective.

Figure 3 provides the visual proof and information on how the venues ended up as anomalies. It shows them across the timeline and by interacting with it like zooming in-out or hovering across the line plots, reveals important insights. Both venues LXE and XDV started off from the beginning of the year but closed down in July maybe due to consistent low visitors. At the same time, venue KLZ started operating from July, and NRB from August. Despite starting mid-way through the year their upward trend indicates they are steadily experiencing visitor growth. Similarly, both MXL and LRT started business from April with upward growth as well. It also reveals that LLJ and KFO are the newest additions from October but despite being the youngest venues they have accumulated lots of visitors and the growth looks rapid within just 2 months. This is the value of interactive plot; it allows to scrutinize data points of interest with great depth.

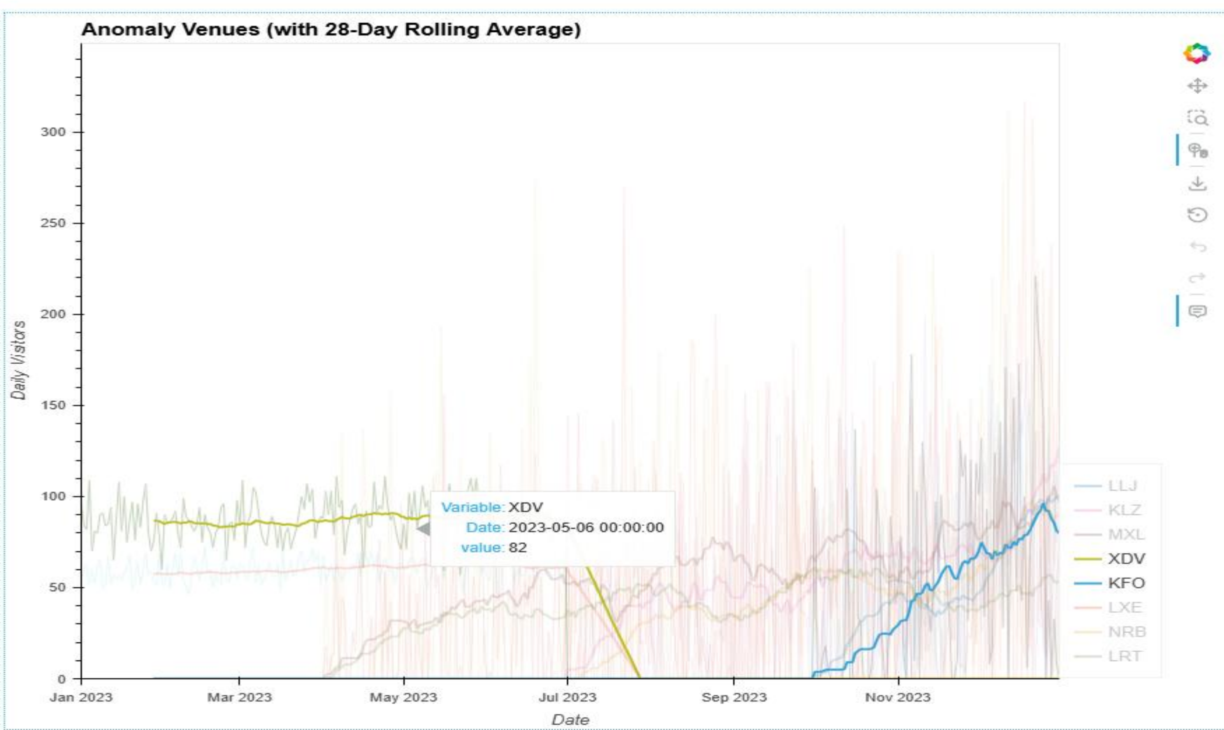

Figure 3: Interactive Plot (legends muted) to explore anomalies

2.4. Interactive Line Plot to Explore High-volume Venues

After brushing up in reporting the anomalies and their exploration, the next step is to focus on the high performers. It may not be the same kind of probing as before, but Figure 4 also allows to explore the high-volume venues in greater depth. Being an interactive plot, it equips the reader with tools for zoom, hover as well as the interactive legend for isolation of the venues. This line plot is also rolling averaged to 28 days to ensure smoothness and remove daily noise, helping to spot trend across the time frame. This visualization reveals whether the high-volume venues are generally growing, declining, or remaining flat over the year, giving ChrisCo insight into overall trajectory. The code identity is same as it was for Figure 3. The adaptation done was also same, just this time it was for high-volume venues.

Figure 4 visualizes the 4 high performing venues PFN, VAU, UOR and VUJ. Key insights that can be extracted from this visual is that UOR is consistently the best performing venue, closely followed by VUJ, VAU and PFN are slightly lower comparatively but they are quite close. This signals that the hierarchy among them is stable as well. In terms of volatility, PFN is very stable with its plot almost looking as if a flat line. UOR shows slight variations throughout the year but in a controlled manner showing steady increase towards the end of the year, on the other hand both VUJ and VAU show moderate fluctuations. PFN's stable nature allows easy forecasting as it is more predictable unlike others. Even with smoothing, there are appearances of gentle waves across the figure, indicating recurring patterns i.e., underlying seasonality.

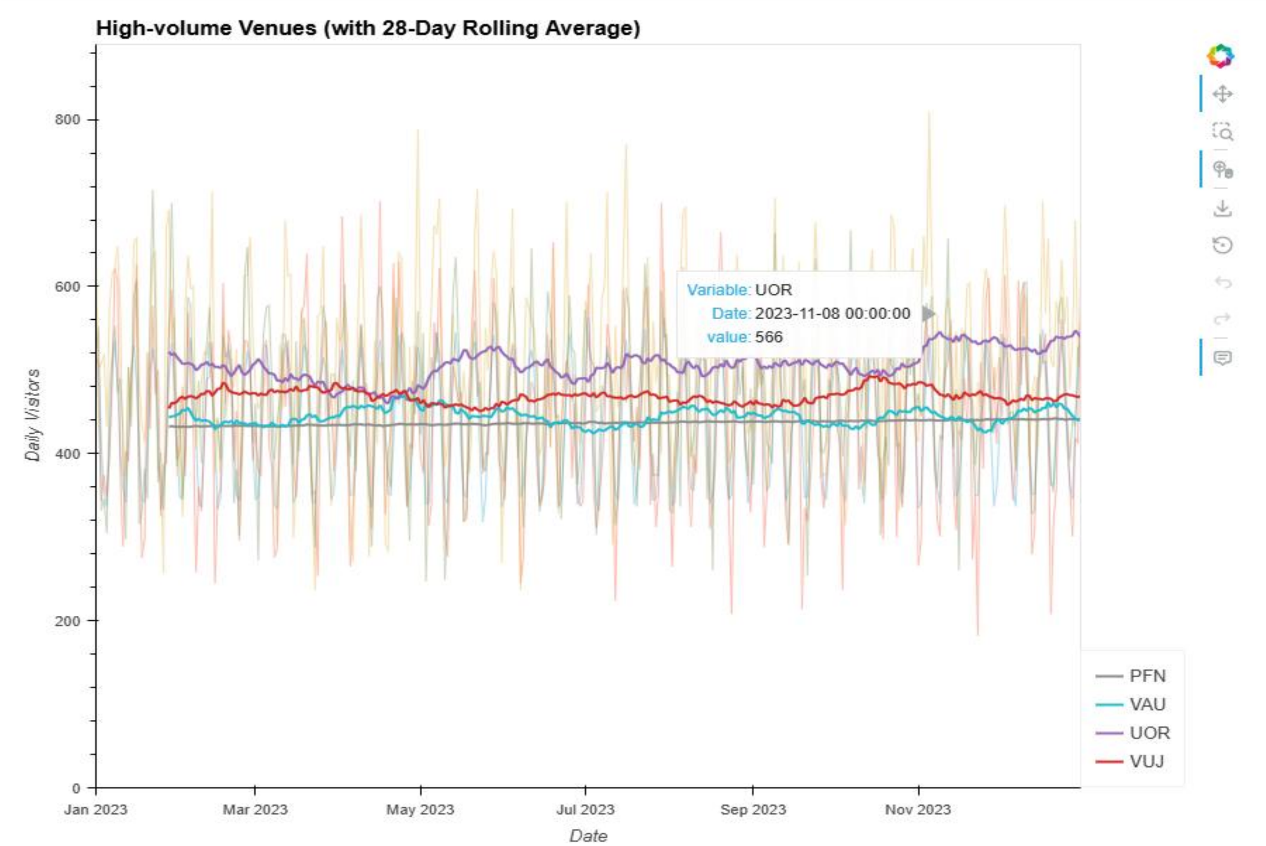

Figure 4: Interactive plot of High-volume venues

2.5. Seasonality Across High-volume and Medium-volume Venues

Next step in the visualization flow is to address seasonality. So far, there has been lots of occurrences in the line plots where the trends over time follow a regular pattern of seasonal peaks and troughs. This periodic behavior of data points is not coincidence it is a pattern which is important to figure out because for good performing venues it can help the company prepare for peak time of crowds so preparations can be done to handle them ensuring sustainable business offering and good services. Autocorrelation is a way to explore seasonality because it measures the correlation of a time-series with itself i.e., how a time series relates to its own past values like a day before, week, month etc. Code identity is from "L05 Distribution/10Autocorrelation all.py". To adapt to this dataset instead of plotting all the venues possibly causing information overload, it was made to plot only for high and medium volume venues.

Figure 5 shows that all 12 high and medium-volume locations have consistent, tight "zigzag" pattern. This rapid oscillation indicates a 7-day cycle which means that the venues have massive, predictable swings between peak days (likely weekends/holidays) and troughs (weekdays). Also, it can be noticed that the spikes decrease gradually instead of abrupt decay showcasing that trends and patterns persist throughout the day meaning if a venue is busy this Sunday it will most likely be busy next Sunday too. Hence, they all indicate consistent and predictable seasonal behavior in visitor demand.

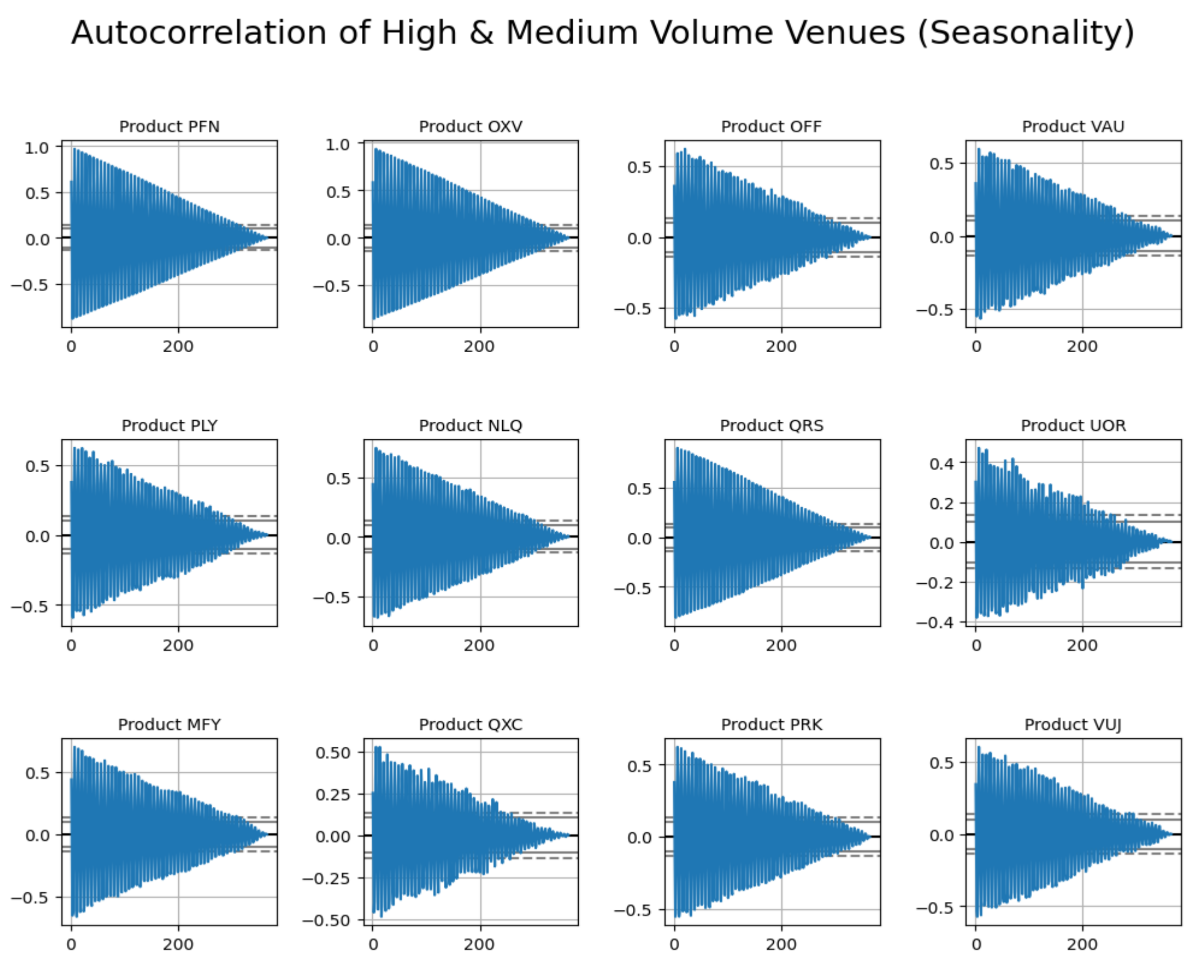

Figure 5: Autocorrelation plot for high and medium-volume venues

2.6. Correlation Among Summary Data for Venues

As correlation has been explored before on the time series data frame of venues, its equally essential to look into possible correlations among different attributes that provide metadata about the venues. This is important because the performance across different venues is heavily dependent on these factors. Computing correlation among the 6 attributes and conveying them all is most efficiently done by heatmap using color intensity to show magnitude of the correlation which immediately helps any reader visually get the message about which pairs are strong(darker) or weak (lighter). Additionally, annotating the correlation values in the heatmap adds quantitative information on top. The code identity is from "L04Correlation/04HeatMap correlation all.py".

In Figure 6, the strongest relationship is between daily visitors and venue distance (0.93) which implies that venues attracting larger crowds also draw visitors from farther distances. Moving on, there is another strong relationship between age of visitors and the spending (0.74). This indicates that venues with older visitors tend to generate higher spending. Maybe an indication to the purchasing power that comes with age. Over the plot, there are several moderate negative correlations present between visitor numbers and visit duration (-0.32) as well as between distance and duration (-0.30), implying that higher traffic and more distant venues experience shorter visit durations from visitors.

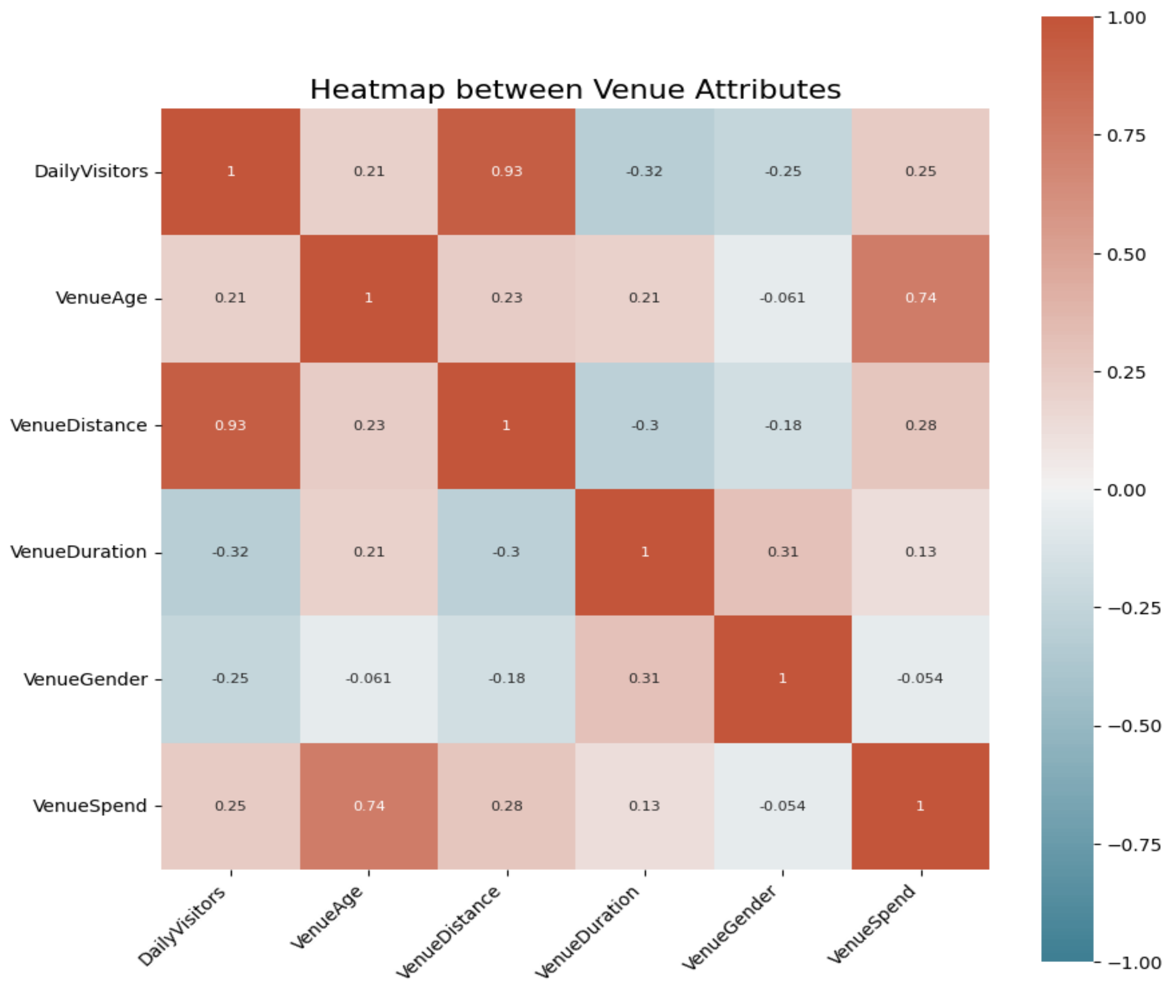

Figure 6: Heatmap of summary data for all venues

2.7. Scatterplot on Daily Visitors vs Venue Distance (Segmented)

Figure 6 not only showed the existence of correlations across different attributes but also quantified them. To build on top of that, the scatterplot visualizes and verifies the correlation among the strongest relationship identified between visitor volume and venue distance attributes. Furthermore, it is segmented with the help of colors to represent different volume venues helping the reader visually identify them. If heatmap was a macro-view of overall attributes, this plot is the microscopic view of the strongest identified correlation. The code identity is from “L04Correlation/14ScatterPlot price vs cost coloured.py. The adaptation for this dataset was minimal, the summary data was segmented for high, medium and low-volume venues.

Figure 7 confirms the correlation by showing a clear positive relationship between daily visitor numbers and the distance travelled by them. Low-volume venues (green) are tightly clustered at lower visitor counts and shorter travel distances, indicating primarily local attendance. Medium-volume venues (orange) occupy a middle range, with both visitors and travel distances increasing. On the other hand, High-volume venues are distinctly separated, attracting huge number of visitors from significantly longer distances. This possibly hints at a hypothesis like what if it is a venue that offers wider or exclusive offerings prompting lots of people even from far to visit it. An interesting discovery is the sighting of an outlier among the high-volume venues. Compared to its peers, despite being high traffic venue it has lower travel distance, possibly hinting at a high local demand or maybe in a densely populated area where demand is from a large local population rather than distant visitors. The well-defined groupings with minimal overlap, reaffirm that visitor volume is strongly associated with how far individuals are willing to travel.

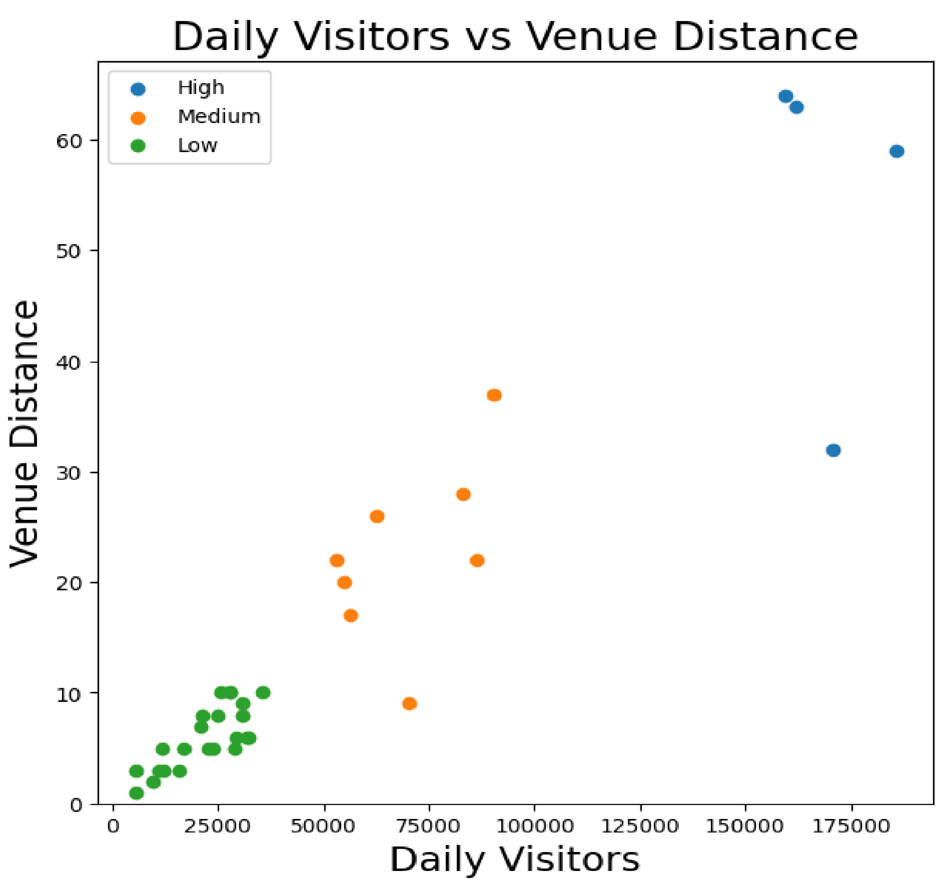

Figure 7: Scatterplot of Daily Visitors vs Venue Distance

2.8. Interactive Bubble Plot of Venue Age vs Venue Spend (vs Daily Visitors)

A notable correlation identified from the heatmap was between visitors age and their spending. This relationship warranted further investigation as it can support the development of meaningful hypotheses and provide insights that the company can use for its growth. To make exploration more effective, an interactive bubble plot proves valuable, as it enables simultaneous visualization of multiple attributes such as customer age, average spend and visitor volume, while allowing deeper examination of individual data points. This interactivity makes it easier to detect patterns, clusters and outliers that may not be apparent in static charts. The zoom, hover and different other interactive tools in this visualization aids in dynamically look into every venue more closely. The code identity is from "L07Interaction/14Bubbleplot marketing sales profit.py, requiring only minor adjustments for this dataset, like adjusting the scaling factor for bubble size.

Figure 8 on a high level confirms the positive relationship between average customer age and average spend across venues conveyed from the heatmap. The implication is that with higher spending is generally associated with older visitor groups, venues with younger audiences tend to cluster at lower spend levels while those with older audiences show a wider range and higher values of average spend. For multidimensional analysis, the size of bubbles is made to represent daily visitor numbers. It indicates that both high and low volume venues exist at different age and spending levels. While it is noticeable that some larger venues appear in mid-to-higher spend ranges, there is no strong visual pattern suggesting that visitor volume is directly linked to either customer age or spending. This is a very important insight because initially looking only at the correlation between age of visitors and spending, a possible hypothesis could have been developed about how age factor influences revenue generation for the venues. Similarly, if there is high spending at a venue, it must surely come due to spending from high volume of visitors. But with the help of this plot, it is revealed that several venues with high average spend have relatively low visitor volumes for example: at Average venue age 55 (x-axis) and average venue spend 36 (y-axis) with the bubble size small, indicating that high spend doesn't necessarily translate from crowded venues. Other different venue demographics are also revealed when hovering over the datapoints, revealing more information about other attributes for each of them. Additionally, venues with moderate visitors age but higher visitor numbers may represent more significant contributors. Hence, it can be concluded that age alone is insufficient for explaining revenue patterns and that visitor volume must be considered as well.

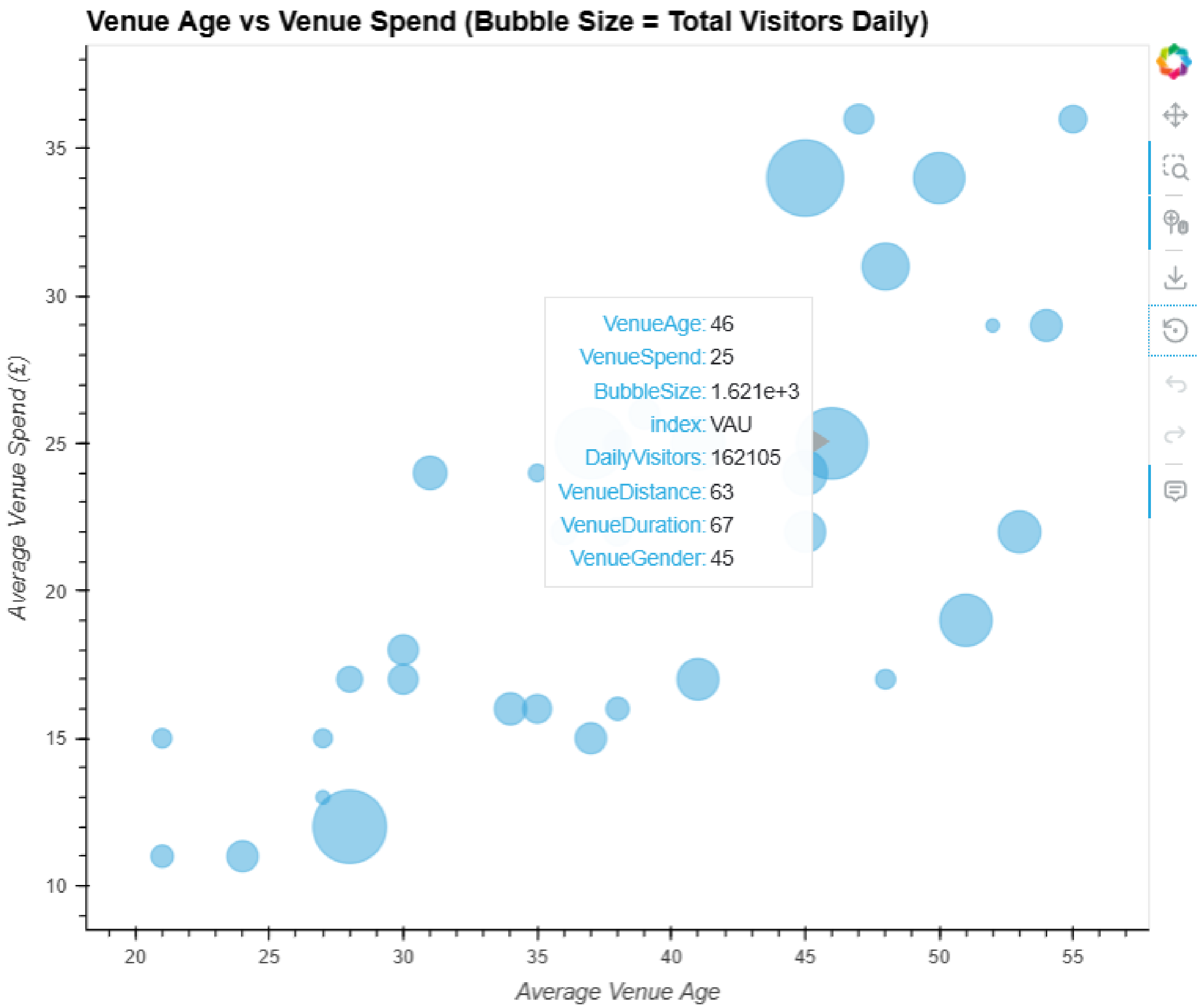

Figure 8: Interactive Bubble plot for Venue Age vs Venue Spend (vs Daily visitors)

3. Critical Review

From the first day of starting this module, I was really fascinated by how the same message can be depicted and conveyed in so many different ways, especially through different visuals and charts. It helped me learn that by using the proper visual and way of communicating it, can leave a deeper impact to the audience. Starting out with the module, I had this quote in my mind that "A picture is worth a thousand words" but when it was finished, I came to slightly disagreeing with it. By studying this module, I believe that indeed a picture may well be worth a thousand words but only if it is the right one. If the picture is not right and cannot speak the correct information than it has no worth. Instead, it can be misinterpreted and wrong conclusions can be drawn.

I learnt how to approach data storytelling and tried my best to apply it in the coursework as well by following a narrative flow. First, I started from a high overview of the data, then segmented it to focus on important details one at a time. I tried to go from macro level to narrower with each step, also allowing interactive engagements using hvplot package that I learnt in this module. Like a story, each new visualization was structured step on top of previous ones guiding the reader from raw to actionable insights and help them reach a certain conclusion and aid decision making.

I have also used best design practices taught in the module keeping visual perception in mind while designing them and minimize cognitive load to the brain by segmenting the data to ensure there is minimal occurrence of information overload, and also the use of pre-attentive attributes where possible so that the readers can grasp what is happening without scratching their head around. Also, most of my visuals follow the principle to minimize eye-travel.

4. Conclusions

After carrying out visual data exploration for ChrisCo, many valuable insights were gathered from the data. These insights are important to consider for the company's well-functioning and growth in the future. Hence, with the help of 8 visualizations across the report, these are the conclusion drawn from each of them:

- The segmented sorted bar chart helped to identify that visitor distribution across venues is highly skewed, with a small number of venues (i.e., UOR, VUJ, VAU, PFN) dominating large share of total visitors while the remaining operate at a significantly lower volume, making segmentation sensible.
- The distribution plot gave an overview on every venue performance showcasing that visitor volume vary significantly, with some notable venues exhibiting right-skewed patterns indicating frequent low-to-moderate attendance with occasional spikes, hinting at possible anomalies within the data.
- Interactive line plots for anomalies helped to investigate why they showed such anomalous patterns with a couple of them closed down mid-year, whereas some opened from mid-year and towards the end of the year accounting for low visitor count.
- Interactive line plots for high-volume venues provided us trajectories of our best performers, which showed stable and predictable patterns with fluctuations year-round.
- The Autocorrelation plot to identify seasonality did its job of revealing strong persistence and clear cyclical patterns across high and medium volume venues, indicating consistent and predictable seasonal behavior in visitor volume.

- The Heatmap of summary data helped in compare correlations across the attributes revealing that visitor numbers are very strongly associated with travel distance.
- The follow up scatterplot verified the strong positive relationship between visitor numbers and travel distance, with clear separation between low, medium and high-volume venues.
- The bubble plot added a new dimension to the analysis, showing that while average spend generally increases with visitor age, visitor volume varies across all age groups, indicating that age alone is not the driving factor for determining overall venue performance.

References

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Tufte, E.R. (2001) The Visual Display of Quantitative Information. 2nd edn. Cheshire, CT: Graphics Press.